

National Exit Exam Questions (UPDATED) (1-77)

ሐምሌ 5፡ 2015 ዓ .ም

July 12, 2023 G.C

Author: Mohammed Nebil

Contact me: +251913828882

@Mohammed_Nebil

1. What will be the output for the Java expression:

$8 - 2 * 5 * (3 + 9 / 3) + 6 / 3$

- A. -150
- B. -30
- C. -50
- D. 50

Answer: C. -50

2. Which one of the following is used the constants in Java?

- A. const
- B. final
- C. define
- D. All of the above

Answer: B. final

3. Which C++ statement is equivalent to short n in variable declaration?

- A. int short n;
- B. unsigned short n;
- C. float short n;
- D. short n;

Answer: B. unsigned short n;

4. Which one of the following is not high-level language?

- A. C++
- B. Java
- C. Python
- D. Assembly

Answer: D. Assembly

5. Which one of the following data structures used for airlines seat reservation?

- A. Array
- B. Linked List
- C. Queue
- D. Stack

Answer: C. Queue

6. Which HTML tag is used for program listing?

- A. <p>
- B. <pre>
- C. <head>
- D. <sub>

Answer: B. <pre>

7. Which phases is used to the existing locks can be released, but no new locks can be acquired?

- A. Expanding phase
- B. Waiting phase
- C. Shrinking phase
- D. All of the above

Answer: C. Shrinking phase

8. Which one of the following exit is an involuntary actions?

- A. Normal exit
- B. Terminate exit
- C. Fatal exit
- D. Error exit

Answer: C. Fatal Exit

9. Which one of the following is wrongly matched?

- A. Fully observable - Partially observable
- B. Discrete - Continuous
- C. Deterministic - Stochastic
- D. Static - Stochastic

Answer: D. Static - Stochastic

10. Which HTML form tag attribute that used maximum number of characters without scrolling or write new line?

- A. MAXLENGTH
- B. TYPE

- C. ACTION
- D. STYLE

Answer: A. MAXLENGTH

11. In which case a dedicated communication path between the sender and receiver has to be established?

- A. Circuit switching
- B. Packet switching
- C. A and B
- D. None of the above

Answer: A. Circuit switching

12. What will be the output of this code?

```
int x = 15, y = 2;  
int z = x/y;  
cout << z;
```

- A. 7.5
- B. 7
- C. 3
- D. 12.5

Answer: B. 7

13. Which one of the following is a type of network topology in which all devices in the network are interconnected?

- A. Star Topology
- B. Ring Topology
- C. Mesh Topology
- D. Tree Topology

Answer: C. Mesh Topology

14. Which of the following generation that stated for the high level languages like C++, Java, Python?

- A. 2nd generation
- B. 3rd generation
- C. 4th generation

D. None of the above

Answer: B. 3rd generation

15. Which of the following generation that stated for interpreted language such like SQL?

A. 4th generation

B. 2nd generation

C. 3rd generation

D. All of the above

Answer: A. 4th generation

16. Which one of the SQL keywords to provide access?

A. REVOKE

B. DISTINCT

C. GRANT

D. None of the above

Answer: C. GRANT

17. Suppose you are network administrator, you want to determine the path between two connections, often a connection to another device will have to go through multiple routers. Which command would you use?

A. ping

B. nslookup

C. traceroute

D. All of the above

Answer: C. traceroute

18. Which one of the following is true about between IPv4 and IPv6?

A. IPv4 is 128-bit and IPv6 is 28-bit

B. IPv4 is physical address

C. IPv6 is logical address

D. IPv4 is 32-bit and IPv6 is 128-bit

Answer: D. IPv4 is 32-bit and IPv6 is 128-bit

19. Which one of the following notation is comfortable for the human programmers?

- A. Infix notation
- B. Prefix notation
- C. Postfix notation
- D. None of the above

Answer: A. Infix notation

20. Which one of the data communication that used for direct transmission?

- A. Induced
- B. Crosstalk
- C. A and B
- D. None of the above

Answer: B. Crosstalk

21. Look at the following fragment code:

```
int p = 10;
int n = 5;
if (n > p) {
    cout << (n+p) << endl;
}
```

Which algorithm property is not satisfied by the above C++ coded algorithm?

- A. Feasibility
- B. Finiteness
- C. Sequential
- D. Efficiency

Answer: D. Efficiency

22. Which one of the following is true regarding constructors in Java?

- A. Constructors are invoked to create objects without using new operator
- B. Constructors can have a return type
- C. A constructor can have different name from its class name
- D. Constructors play the role of initializing objects

Answer: D. Constructors play the role of initializing objects

23. Identifying and describing the fundamental software system abstractions and their relationships. Establishing the overall system architecture in which phase these activities are done?

- A. System design
- B. Requirement Analysis and definition
- C. System testing
- D. Implementation

Answer: A. System design

24. _____ is a set of procedures that uses representational language to conclude new fact from the existing ones.

- A. Inference Engine
- B. Explanatory facility
- C. Interface
- D. Knowledge Base

Answer: A. Inference Engine

25. What will be the value of $|Z|_d$ if it is given that string $Z = \text{caddba}$?

- A. 3
- B. 4
- C. 2
- D. 12

Answer: B. 4

26. Which one of the following is a type of compiler which compiles only the changed lines from source code and updates the object code accordingly?

- A. Retargetable compiler
- B. One pass compiler
- C. Incremental compiler
- D. Parallelizing compiler

Answer: C. Incremental compiler

27. What is the time complexity order of push() operation of Stack data structure implemented using array?

- A. $O(n)$
- B. $O(\log_2 n)$
- C. $O(n^2)$
- D. $O(1)$

Answer: $O(1)$

28. Which one of the following is uninformed searching method?

- A. A^* search
- B. Iterative improvement
- C. Breadth first search
- D. Greedy search

Answer: C. Breadth First Search

29. Which one of the following is a false reason for why we use HTML?

- A. Because it works with all platforms, all browsers and all web servers
- B. Because it is quickly transferred over a network.
- C. Because HTML files are flat text and very small
- D. Because it is proprietary technology

Answer: Because it is proprietary technology

30. Writing comments in between our C++ program makes our program _____

- A. Portable
- B. Maintainable
- C. Reliable
- D. Efficient

Answer: B. Maintainable

31. Given input alphabet $\Sigma = \{a, b, c, d\}$ and empty string λ . Then, which one of the following is false about λ ?

- A. $\lambda ab = ab$
- B. For any string Z , $Z^0 = \lambda$
- C. λ is an element of Σ^+
- D. $\lambda = 0$

Answer: D. $\lambda = 0$

32. What will be the value of Z^R if it is given that string $Z = \text{caddba}$?

- A. abddac
- B. cad
- C. abd
- D. caddba

Answer: A. abddac

33. Which one is a space where an agent perceives and acts so that an intelligent agent designer has to understand the type of it?

- A. Environment
- B. Actuator
- C. Sensor
- D. Search space

Answer: A. Environment

34. What is the output of the following fragment JavaScript?

```
<script>
  for (n = 1; n < 7; n++) {
    if (n % 5 == 0) {
      document.write(n + " is multiple of 5");
    }
  }
</script>
```

- A. 5 is multiple of 5
- B. 1 2 3 4 5 is multiple of 5 6
- C. 5 is multiple of 5 6 7
- D. 1 2 3 4 5 6

Answer: A. 5 is multiple of 5

35. If we have two strings $Z = \text{cddc}$ and $W = \text{ababb}$, then the concatenation of the two strings ZW will be?

- A. $ZW = \text{cddcababb}$
- B. $ZW = \text{cdab}$

C. ZW = ababbcdde

D. ZW = ca

Answer: A. ZW = cddcababb

36. XYZ Company is attacked by intruders which causing system crashes, slowing down of the services and denying services to targeted users. Therefore, the company is attacked by?

A. Access attack

B. Viruses

C. Information gathering

D. Denial of services (DoS)

Answer: D. Denial of services (DoS)

37. Which performance metrics can be used to measure the time taken to perform a read or write operation?

A. Transfer rate

B. Memory cycle time

C. Fetching time

D. Access time

Answer: D. Access time

38. Which protocol of the TCP/IP is a connectionless protocol?

A. User data protocol

B. Internet protocol

C. Transmission protocol

D. The OSI reference protocol

Answer: A. User data protocol

39. Which one is true about the following JavaScript statement?

```
document.write("Hello World");
```

A. write is a property

B. write is a method

C. document is a property

D. write is an object

Answer: B. write is a method

40. Which one of the following is false about scripting languages?

- A. They are platform dependent
- B. They are slower than other compiled codes
- C. They do not need compilers and development environments
- D. They have simple execution model

Answer: A. They are platform dependent

41. If you want to make methods and data fields of classes accessible from any class in any package, which access modifier is appropriate?

- A. protected
- B. default
- C. public
- D. private

Answer: C. public

42. Which one of the following is a voluntary termination of a process?

- A. Error exit
- B. Killed by another process
- C. Terminated by another process
- D. Fatal exit

Answer: A. Error exit

43. Which one of the following wrongly coupled about passive (offline) and active (online) attacks?

- A. Active attack->add or delete messages
- B. Active attack -> traffic analysis
- C. Passive attack -> monitoring traffic
- D. Passive attack -> obtain message contents

Answer: B. Active attack -> traffic analysis

44. Which one of the following is not correct about symmetric key?

- A. It requires sender and receiver to share common secret key
- B. Previous key cannot be used to a new key

- C. Trusted third party can generate and deliver keys
- D. Repetitive key change is encouraged to limit data being compromised

45. Which method can resolve collision in hashed files by proceeding from the occupied position specified by the hash address, the program checks the subsequent positions in order until an empty position is found?

- A. Open addressing
- B. Multiple hashing
- C. Chaining
- D. Cluster indexing

Answer: A. Open addressing

46. Which one doesn't describe loopback address?

- A. It helps to refer to the host itself
- B. It allows programs to access network protocols for local services
- C. It is an external pseudo address
- D. It is denoted by an IP address: 127.0.0.1

Answer: C. It is an external pseudo address

47. Which value of form's METHOD attribute causes a form's contents to be parsed one element at a time

- A. GET
- B. INPUT
- C. ACTION
- D. POST

Answer: D. POST

48. Which one of software testing is most time consuming test phase?

- A. Unit testing
- B. Acceptance testing
- C. System testing
- D. Integration testing

Answer: C. System testing

49. Which technique of code optimization shifts computations from run time to compile time by evaluating constant expressions at compile time and replace the constant expressions by their values?

- A. Copy propagation
- B. Compile time evaluation
- C. Code motion
- D. Dead code Elimination

Answer: B. Compile time evaluation

50. Which one of the following is not correct about function overloading in C++?

- A. Two or more overloaded functions may differ in their number of parameters
- B. Two or more overloaded functions may differ in data types of their parameters
- C. Two or more overloaded functions should have the same function name
- D. Two or more overloaded functions may differ only in their return types

Answer: D. Two or more overloaded functions may differ only in their return types

51. As compared to IPv4, IPv6 can:

- A. Handle more addresses
- B. Use 128-bits
- C. Route slower and memory intensive
- D. Support encryptions

Answer: A. Handle more addresses

52. Which one of the following searching strategies best explains the following statement? "The searching agent is based on problem specific knowledge and the agent knows the distance different state is far away from the goal".

- A. Blind search
- B. Informed search
- C. Uninformed search
- D. Not informed search

Answer: B. Informed search

53. Which one of the following is not correct about heap files?

- A. Sorting is required to read records in particular field order
- B. Linear search can be used to search necessary record

- C. Record insertion is efficient
- D. New records are inserted at the beginning of the file

Answer: D. New records are inserted at the beginning of the file

54. In contrary to the traditional file based approach, in the database approach:
- A. Programs define and access their own data
 - B. Store data item in multiple copies
 - C. Don't care about the definition of programs and data
 - D. Data and programs are defined independently

Answer: D. Data and programs are defined independently

55. Assume you are a database designer and you are given a task to specify entities, relationships, constraints on relations and data types for the data to be stored in the database. At which level of the three-schema architecture would this be addressed?
- A. View level
 - B. Internal level
 - C. External level
 - D. Conceptual level

Answer: D. Conceptual level

56. Which one of the following is correct?
- A. In single user systems many users can login simultaneously
 - B. Single user systems can allow concurrent access to the database
 - C. Multiuser system can allow at most one user at a time
 - D. In Multiuser system many users can access the system concurrently

Answer: D. In Multiuser system many users can access the system concurrently

57. Which one of the following is false about divide and conquer or modularization process?
- A. Separate people can work on each sub-divided part
 - B. Opportunities arise for making the component reusable
 - C. Each individual component is smaller, and therefore easier to understand
 - D. Replacing or changing a component is difficult because it affects other parts extremely.

Answer: D. Replacing or changing a component is difficult because it affects other parts extremely.

58. Which one is true about Minimum Spanning Trees of graphs?

- A. Minimum spanning tree contains all vertices from an input graph
- B. We can have only one unique minimum spanning tree for a graph
- C. Minimum spanning trees are graphs
- D. Minimum spanning trees form cycle

Answer: B. We can have only one unique minimum spanning tree for a graph

59. Assume you are using Hash Table data structure having array size of 6 and your hash function is defined as follows: $h(\text{key}) = \text{key} \% m$, where m is size of your hash table.

Where $\text{key} = 36$ will be stored in your hash table?

- A. At index 35
- B. At index 1
- C. At index 6
- D. At index 0

Answer: D. At index 0

60. Which is not correct about computers evolution?

- A. Increasing component size
- B. Increasing processor speed
- C. Increasing I/O capacity
- D. Increasing memory size

Answer: A. Increasing component size

61. Which one of the following statement is not correct about threads and processes?

- A. Both can create children
- B. Threads within a process execute sequentially
- C. Processes are not independent of one another
- D. If one thread is blocked, another thread can run

Answer: C. Processes are not independent of one another

62. Which one of the following is not correct about system uniformity?

- A. Non-uniform configuration minimizes a probable loss

- B. Uniform configuration increases predictability
- C. In uniform configuration, if some components are poor it doesn't mean the rest are
- D. Non-uniform configuration avoids putting all the resources to the same problem

Answer: C. In uniform configuration, if some components are poor it doesn't mean the rest are

63. Which one of the following is not a wireless attack?

- A. bluesnarfing
- B. phishing
- C. rogue access point
- D. bluejacking

Answer: B. phishing

64. Which access method is used in magnetic disks?

- A. Associative
- B. Sequential
- C. Random
- D. Direct

Answer: D. Direct

65. Which phase of software development process refers to modification of software to reflect changing customer and market requirements?

- A. Development
- B. Evolution
- C. Validation
- D. Specification

Answer: B. Evolution

66. Which one of the following is not correct about Von Newmann's architecture?

- A. Data and instructions are stored separately
- B. Modern computers are based on Von Newmann's architecture
- C. Instruction execution is sequential
- D. Memory content is located by its address

Answer: A. Data and instructions are stored separately

67. If power went off while processing transaction, what should the DBMS do to reverse any changes made to the database

- A. Read_transaction
- B. Fail transaction
- C. Commit transaction
- D. Rollback

Answer: D. Rollback

68. Instead of delivering the system as a single delivery, the development and delivery is broken down into phases with each phase delivering part of the required functionality.

What type of development mode is this?

- A. Incremental
- B. Waterfall
- C. Spiral
- D. Evolutionary

Answer: A. Incremental

69. We should not use Waterfall software development process mode in situations like

- A. Plenty resources with required expertise are available freely
- B. There are no ambiguous requirements
- C. The project is short
- D. Software requirements change quickly

Answer: D. Software requirements change quickly

70. Which one of the following is true about tree data structure?

- A. Tree is static data structure
- B. Tree is linear data structure
- C. Tree forms cycle. In other words tree is cyclic data structure
- D. There is only one unique path when we traverse from one node to another node in a tree

Answer: D. There is only one unique path when we traverse from one node to another node in a tree

71. When a process executes code that possibly changes the value of shared data, its execution has to be mutually exclusive. Therefore, the process is in:

- A. Deadlock
- B. Terminated
- C. Critical section
- D. Blocked

Answer: C. Critical section

72. Which characteristic of requirements specify that the requirement should be phrased so that there is one and only one interpretation for it?

- A. Unambiguous
- B. Correct
- C. Feasible
- D. Verifiable

Answer: A. Unambiguous

73. Which instruction type is used to test the value of a data word or status of composition?

- A. Data processing instruction
- B. Test instruction
- C. Data movement instruction
- D. Branch instruction

Answer: B. Test instruction

74. Which one of the following is false about any language L ?

- A. $L^3 = L^1 L^1 L^1$
- B. $L^0 = L$
- C. $L^* = L^0 UL^1 UL^2 UL^3 UL^4 U$
- D. $L^+ = L^1 UL^2 UL^3 UL^4 UL^5 U$

Answer: D. $L^+ = L^1 UL^2 UL^3 UL^4 UL^5 U$

75. Assume it is given that $f(n) = 4n^2$, $g(n) = 8n$, then which asymptotic notation is correct?

- A. $f(n) = \text{Big-Oh}(g(n))$
- B. $f(n) = \text{Little-Oh}(g(n))$
- C. $f(n) = \text{Big-Omega}(g(n))$

D. $f(n) = \Theta(g(n))$

Answer: C. $f(n) = \Omega(g(n))$

76. Given input alphabet $\Sigma = \{a, b, c, d\}$ which one of the following is an element of set Σ^2 ?

A. b

B. c

C. ac

D. aabbccdd

Answer: D. aabbccdd

77. What is the output of the following C++ fragment code?

```
int data[10] = {5,10,15,20,25,30,35,40,45,50};
int i = 0;
while (i <= 10) {
    if(data[i] % 10 == 0) {
        cout << data[i] << " ";
        i = i + 2;
    }
}
```

A. 5 15 25 35 45 55

B. 10 20 30 40 50

C. 5 10 15 20 25 30 35 40 45 50

D. No output

Answer: D. No output

THANK YOU!